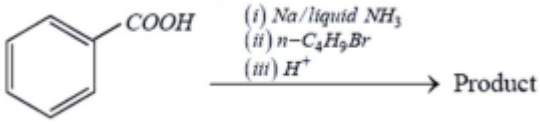
Quizrr Chapter-Wise Test for JEE Advanced - 2025

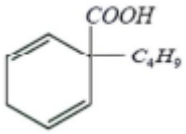
By: C I P H Ξ R

Q.1

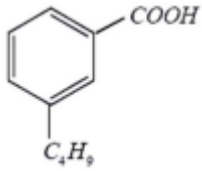
The major product formed in the following reaction is



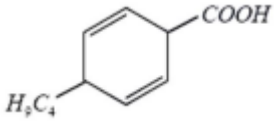
(A)



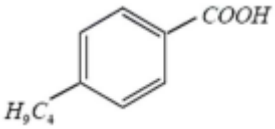
(B)



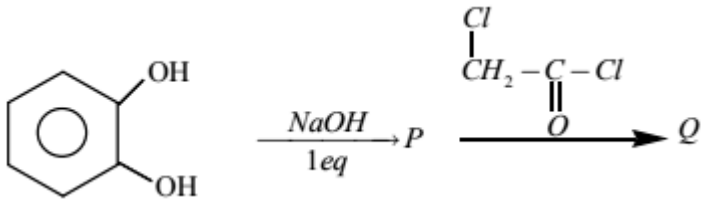
(C)



(D)

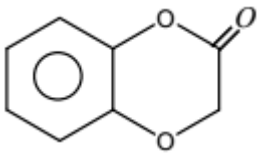


Q.2

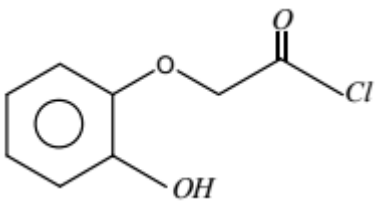


Regarding ' Q ' select the correct statement

(A) ' Q ' is

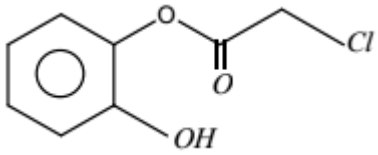


(B) ' Q ' is



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(C) 'Q' is



(D) It's formation is involved carbocation

Q.3

An organic compound contains $\mathrm{C} = 40\%, \mathrm{H} = 6.7\%$ and $\mathrm{O} = 53\%$. Its V.D. is 15. It reduces fehling's solution. A is -

(A) Acetaldehyde

(B) Formaldehyde

(C) Formic acid

(D) Glucose

Q.4

Minimum pH of one molar aqueous solution will be shown by the compound -

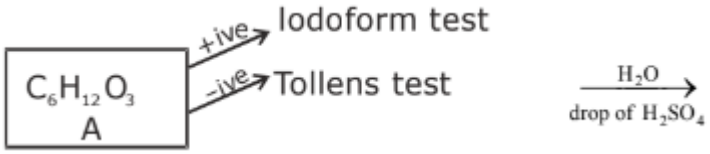
(A) $\mathrm{ClCH_2COOH}$

(B) $\mathrm{BrCH_2COOH}$

(C) $\mathrm{CH_3CH_2COOH}$

(D) $\mathrm{CH_3COOH}$

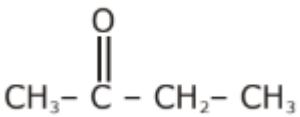
Q.5



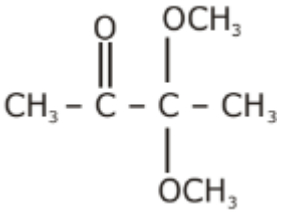
Positive Tollen's Test

Compound (A) is

(A)

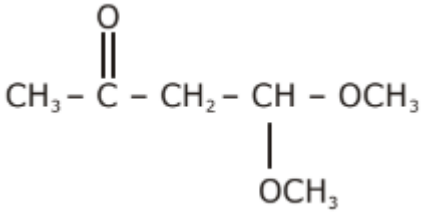


(B)

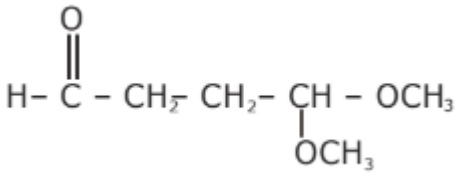


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(C)



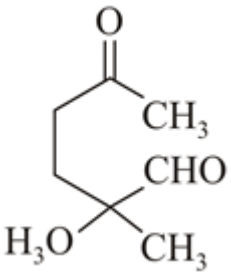
(D)



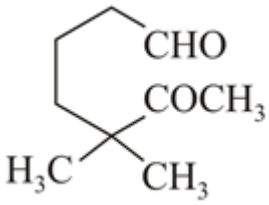
Q.6

$Me_2CHCHO + CH_2 = CHCOCH_3 \xrightarrow[(ii)OH^-]{(i)\text{Michael addition}} [X]$ product is

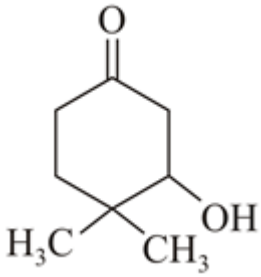
(A)



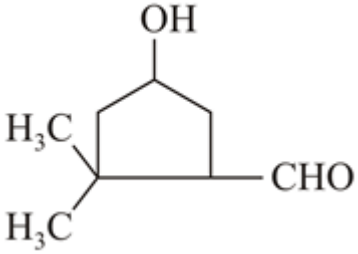
(B)



(C)

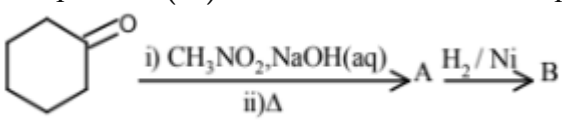


(D)

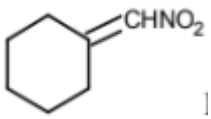


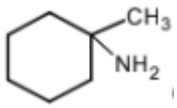
Q.7

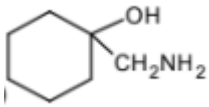
The product (B) formed in the reaction sequence

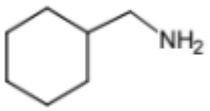


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(A) 

(B) 

(C) 

(D) 

Q.8

Pyrolignious acid obtained by destructive distillation of wood, can be used for the manufacture of the following by passing through $Ca(OH)_2$ –

(A) Methyl alcohol

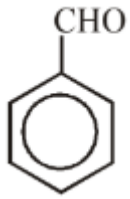
(B) Acetaldehyde

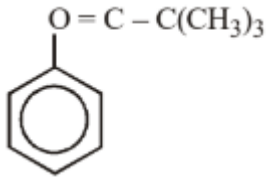
(C) Acetic acid

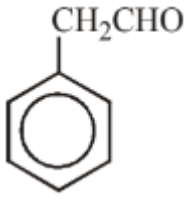
(D) Propanol

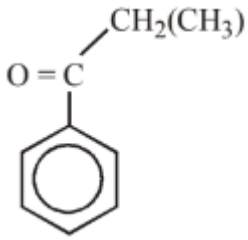
Q.9

Which of the following molecule(s) contain deuterium after reaction with dil. $NaOD$ in D_2O ?

(A) 

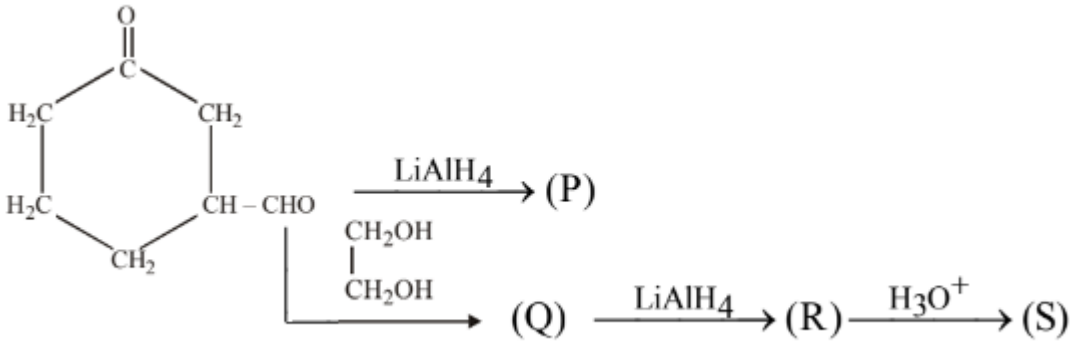
(B) 

(C) 

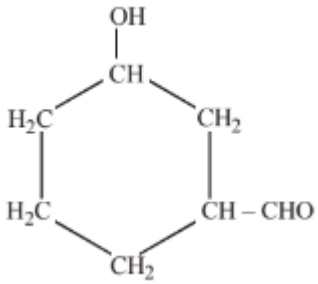
(D) 

Q.10

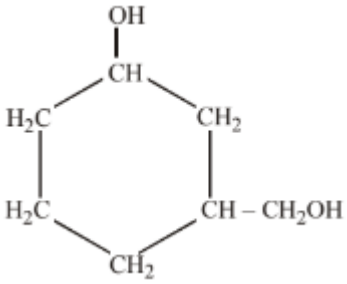
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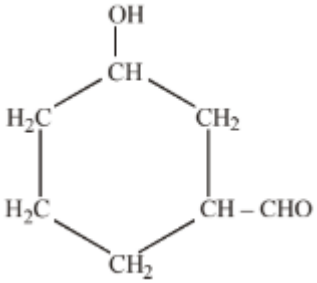
(A) Structure Q is



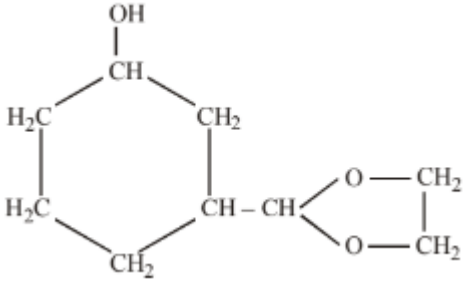
(B) Structure P is



(C) Structure S is

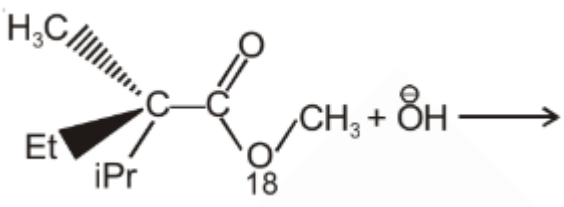


(D) Structure R is



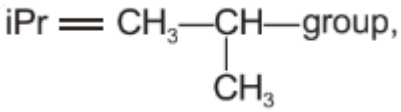
Q.11

Which of the following statements are correct regarding given reaction?



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[where



O^{18} = represents labelled oxygen]

(A) The given reaction primarily follows S_N2 mechanism

(B) In the given reaction condition inversion occurs at the chiral carbon

(C) Labelled ^{18}O is present in formed carboxylic acid

(D) The given reaction primarily follows addition / elimination mechanism

Q.12

Passage:

Novocaine, a local anaesthatic, is a compound with molecular formula, $C_{13}H_{20}O_2\ N_2$. It is insoluble in water and dil. $NaOH$, but soluble in dil. HCl. Upon treatment with $NaNO_2$ and HCl and then with β-naphthol, a highly coloured solid is formed. When Novocaine is boiled with aqueous $NaOH$, it slowly dissolves. The alkaline solution is shaken with ether and layers are separated.

Acidification of the aqueous layer causes the precipitation of white solid A, continued addition of acid causes A to redissolve. Upon isolation A is found to have melting point of $185 - 186^\circ C$ and the formula $C_7H_7O_2\ N$. Evaporation of ether layer leaves a liquid B of formula $C_6H_{15}ON$.B dissolves in water to give a solution that turns litmus blue. Treatment of B with acetic anhydride gives C, $C_8H_{17}O_2\ N$ which is insoluble in water and dilute bases, but soluble in dilute HCl.

B is found to be identical to the compound formed by the action of diethylamine on ethylene oxide.

Question:

The molecule C is functionally -

(A) amino acetate

(B) nitro acetate

(C) cino acetate

(D) diazo salt

Q.13

Passage:

Novocaine, a local anaesthatic, is a compound with molecular formula, $C_{13}H_{20}O_2\ N_2$. It is insoluble in water and dil. $NaOH$, but soluble in dil. HCl. Upon treatment with $NaNO_2$ and HCl and then with β-naphthol, a highly coloured solid is formed. When Novocaine is boiled with aqueous $NaOH$, it slowly dissolves. The alkaline solution is shaken with ether and layers are separated.

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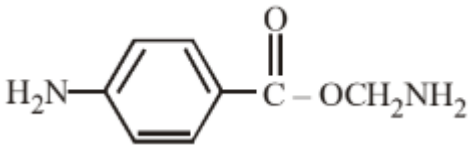
dilute bases, but soluble in dilute HCl.

B is found to be identical to the compound formed by the action of diethylamine on ethylene oxide.

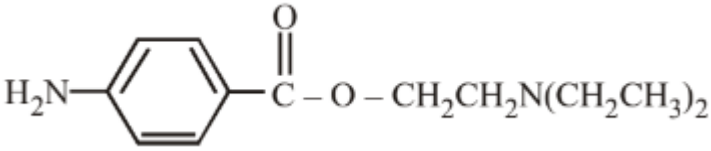
Question:

When compound B is treated with p-nitrobenzoyl chloride followed by moderate reduction with Ni/H_2, the compound formed is —

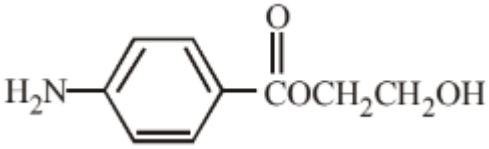
(A)



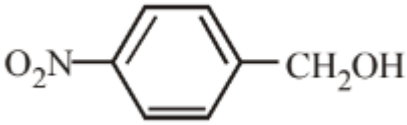
(B)



(C)



(D)



Q.14

Passage:

Novocaine, a local anaesthatic, is a compound with molecular formula, $C_{13}H_{20}O_2\ N_2$. It is insoluble in water and dil. $NaOH$, but soluble in dil. HCl. Upon treatment with $NaNO_2$ and HCl and then with β-naphthol, a highly coloured solid is formed. When Novocaine is boiled with aqueous $NaOH$, it slowly dissolves. The alkaline solution is shaken with ether and layers are separated.

Acidification of the aqueous layer causes the precipitation of white solid A, continued addition of acid causes A to redissolve. Upon isolation A is found to have melting point of $185 - 186^\circ C$ and the formula $C_7H_7O_2\ N$. Evaporation of ether layer leaves a liquid B of formula $C_6H_{15}ON$.B dissolves in water to give a solution that turns litmus blue. Treatment of B with acetic anhydride gives C, $C_8H_{17}O_2\ N$ which is insoluble in water and dilute bases, but soluble in dilute HCl.

B is found to be identical to the compound formed by the action of diethylamine on ethylene oxide.

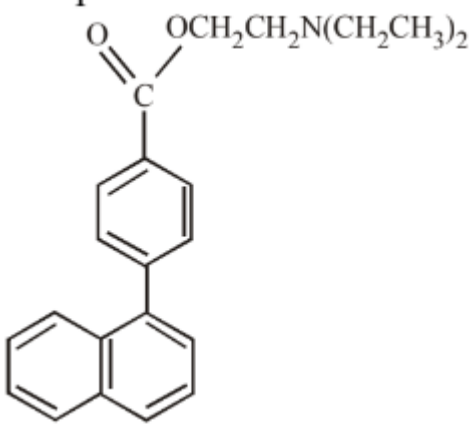
Question:

$$\text{Novocaine} \xrightarrow{NaNO_2.HCl} \xrightarrow{\beta-\text{naphthol}} \text{Product.}$$

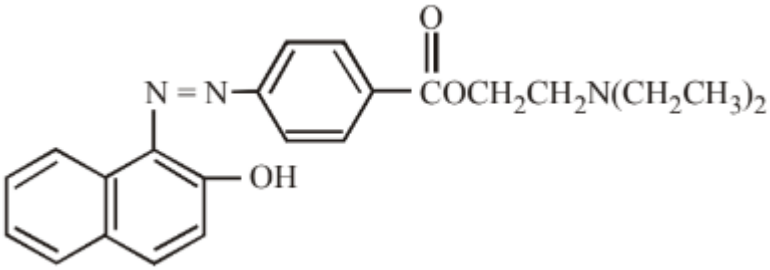
The product is -

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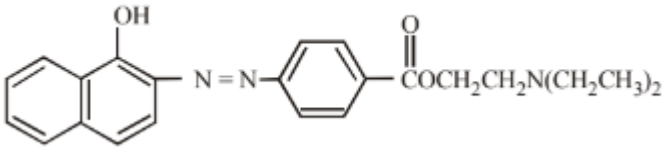
(A)



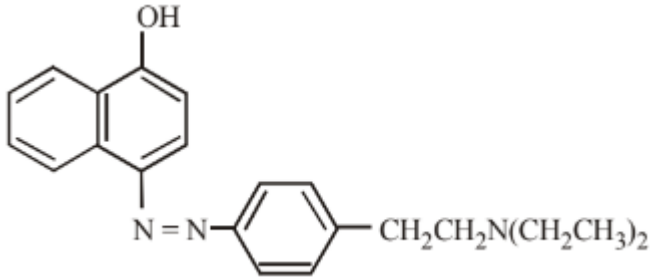
(B)



(C)



(D)



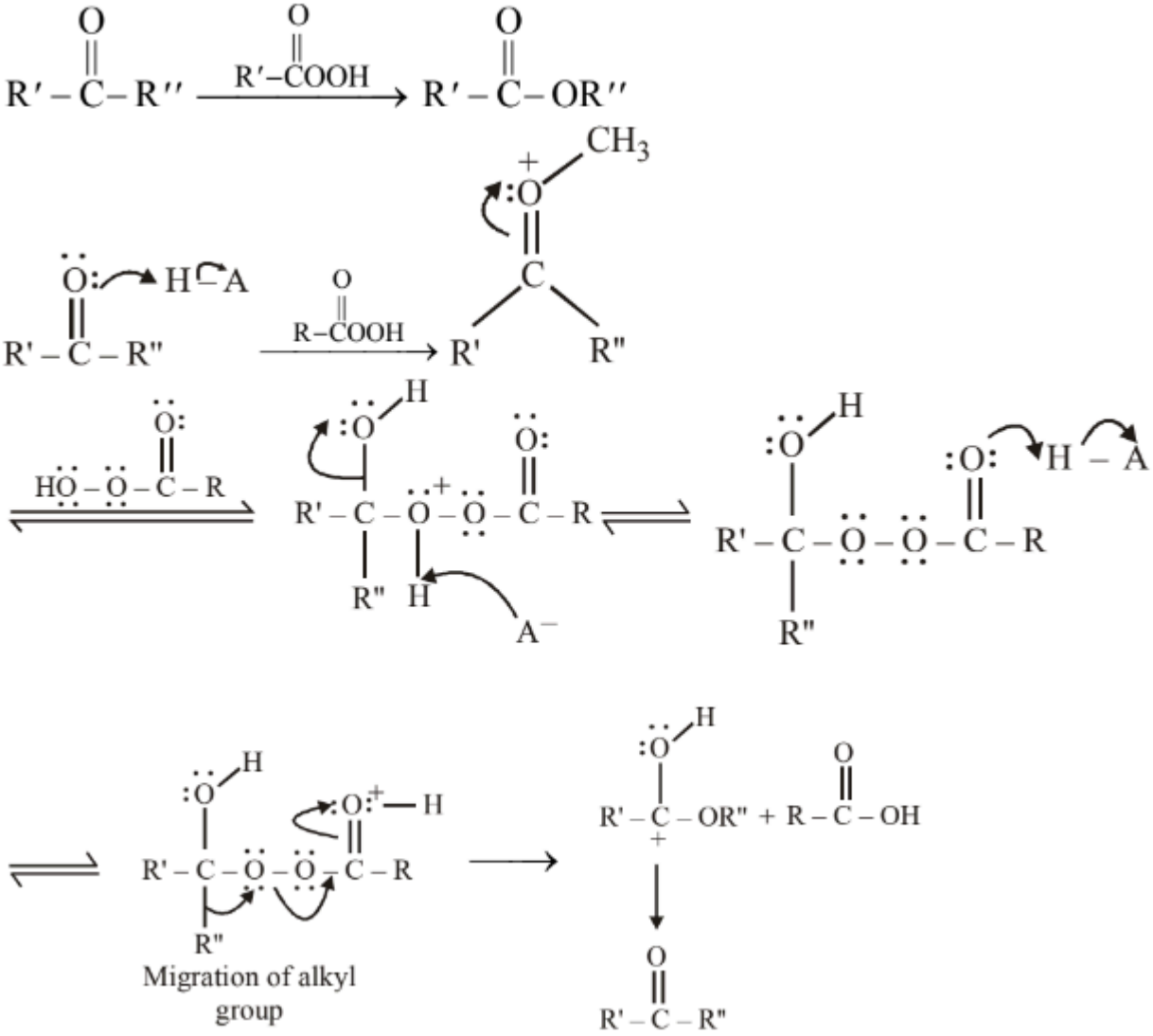
Q.15

Passage:

The Baeyer-Villiger oxidation is a useful method for conversion of carbonyl compounds to esters by the insertion of

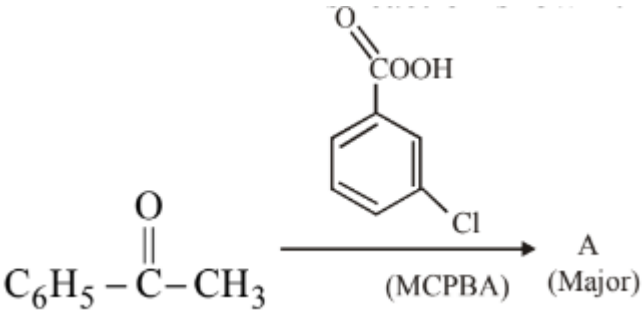
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an oxygen atom from a peroxycarboxylic acid.



The nature of the product obtained in this reaction depends up on the migratory aptitude of alkyl group. Studies of this reaction shown that migratory aptitude of groups is $H >$ phenyl $> 3^{\circ}$ alkyl $> 1^{\circ}$ alkyl $>$ methyl

Question:



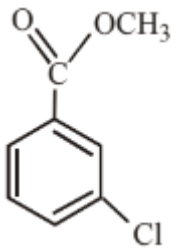
In this reaction A is -

(A) $C_6H_5-\overset{\overset{O}{\|}}{C}-OCH_3$

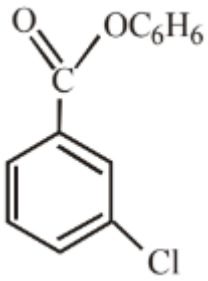
(B) $CH_3-\overset{\overset{O}{\|}}{C}-OC_6H_5$

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(C)



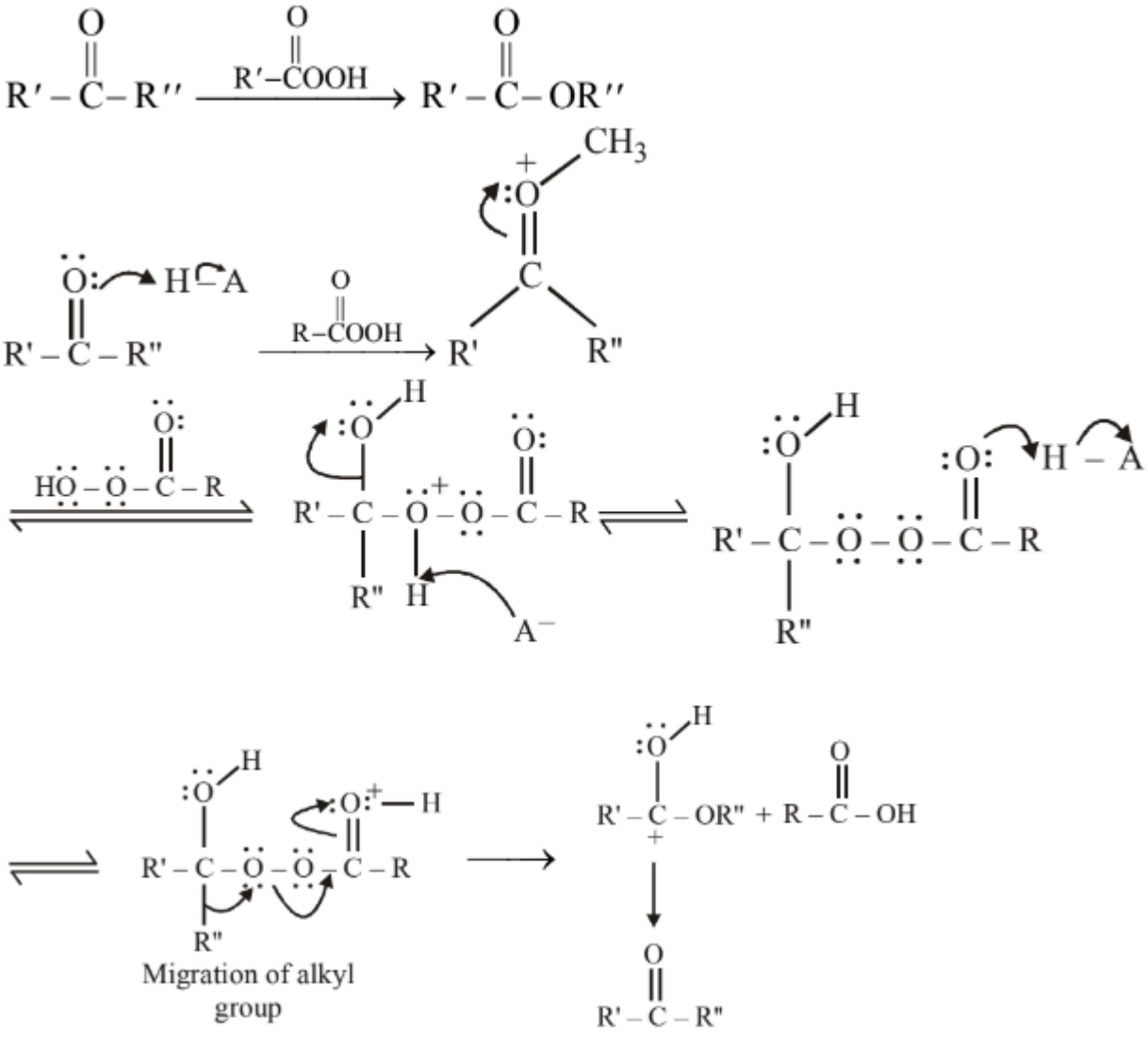
(D)



Q.16

Passage:

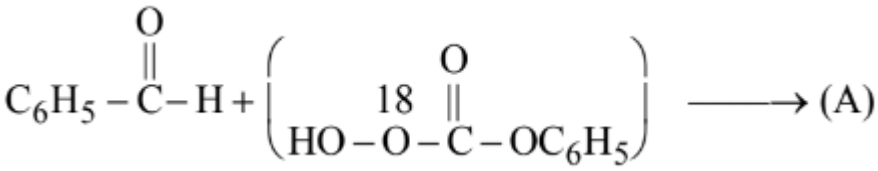
The Baeyer-Villiger oxidation is a useful method for conversion of carbonyl compounds to esters by the insertion of an oxygen atom from a peroxycarboxylic acid.



The nature of the product obtained in this reaction depends up on the migratory aptitude of alkyl group. Studies of this reaction shown that migratory aptitude of groups is H > phenyl > 3° alkyl > 1° alkyl > methyl

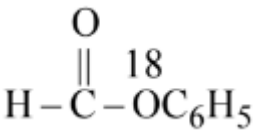
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Question:

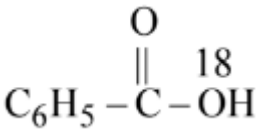


In this reaction is -

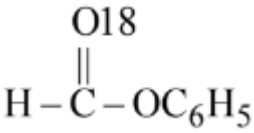
(A)



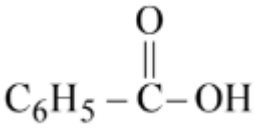
(B)



(C)



(D)



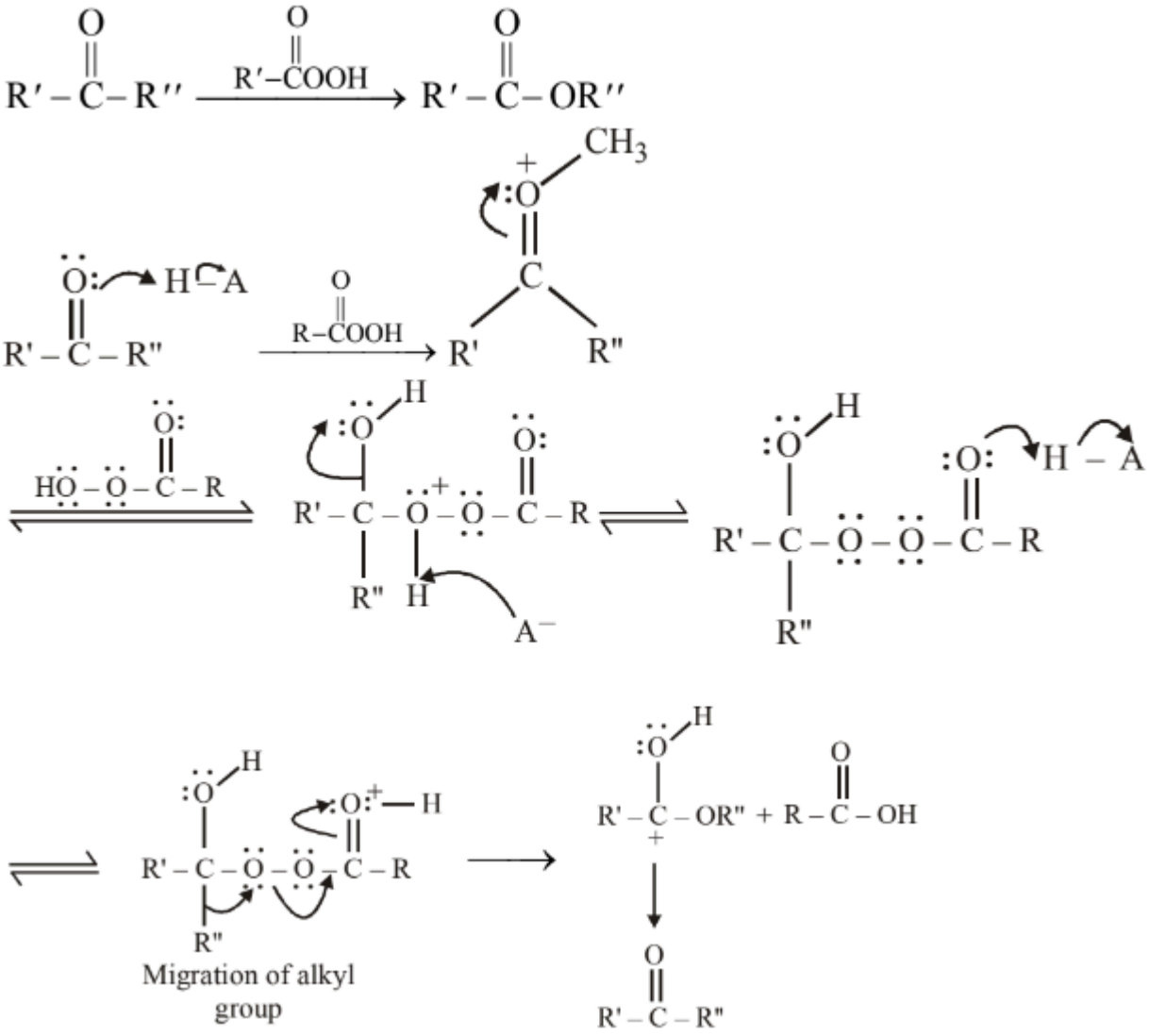
Q.17

Passage:

The Baeyer-Villiger oxidation is a useful method for conversion of carbonyl compounds to esters by the insertion of

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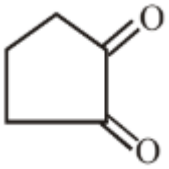
an oxygen atom from a peroxycarboxylic acid.

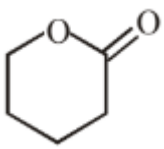


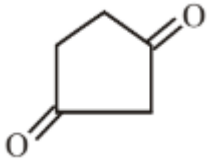
The nature of the product obtained in this reaction depends up on the migratory aptitude of alkyl group. Studies of this reaction shown that migratory aptitude of groups is $H >$ phenyl $> 3^{\circ}$ alkyl $> 1^{\circ}$ alkyl $>$ methyl

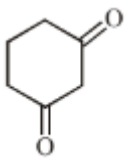
Question:

Cyclopentanone on Baeyer-Villiger oxidation gives -

(A) 

(B) 

(C) 

(D) 

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Q.18

Match the following :

Column I **(Hydroxy acids which are being heated)**	**Column II** **(Characteristics of the product formed)**
(A) $HOCH_2-COOH$	(p) Lactides
(B) $Ph-\underset{\underset{OH}{|}}{CH}-COOH$	(q) Cyclic
(C) $Ph-\underset{\underset{OH}{|}}{CH}-CH_2COOH$	(r) Exhibits geometrical isomerism
(D) $Ph-\underset{\underset{OH}{|}}{CH}-CH_2CH_2COOH$	(s) Lactones
	(t) Can be optically active

(A) (A) - p,q, (B) - p,q,r,t, (C) - r, (D) - q,s,t,

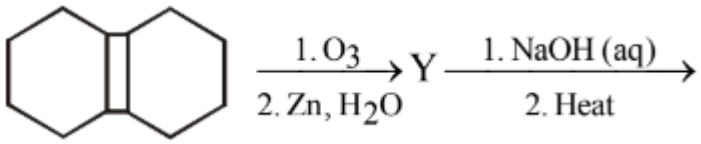
(B) (A) - q, (B) - p,q,s, (C) - p,r, (D) - p,q

(C) (A) - r,t, (B) - p,r,t, (C) - r,t, (D) - r

(D) (A) - s, (B) - p,s,t, (C) - p,r, (D) - r,s,t

Q.19

In the scheme given below, the total number of intramolecular Aldol condensation products formed from 'Y' is:



Q.20

0.535 g ethanol and acetaldehyde mixture when heated with Fehling solution gave 0.12 g of a red ppt. What is the percentage of acetaldehyde in the mixture (round off to nearest integer).
(At weight of $Cu = 63.8$)

Answers & Solutions

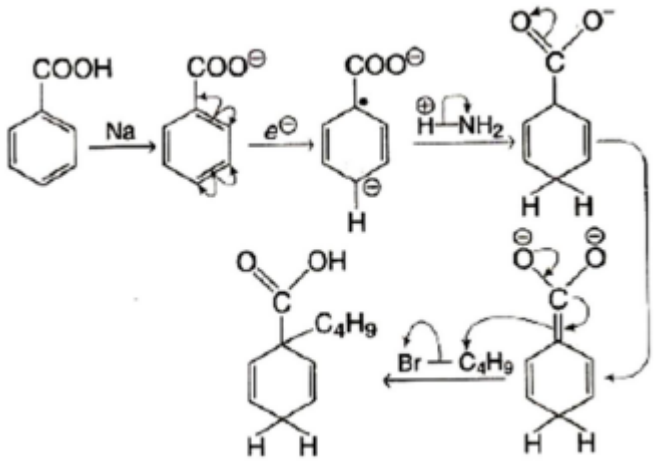
Q.1 Answer:

—

Solution:

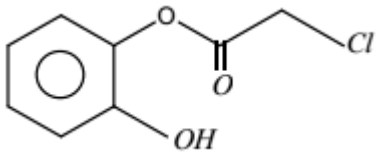
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$Na + NH_3 \longrightarrow Na^+ + e^- [NH_3]_n$

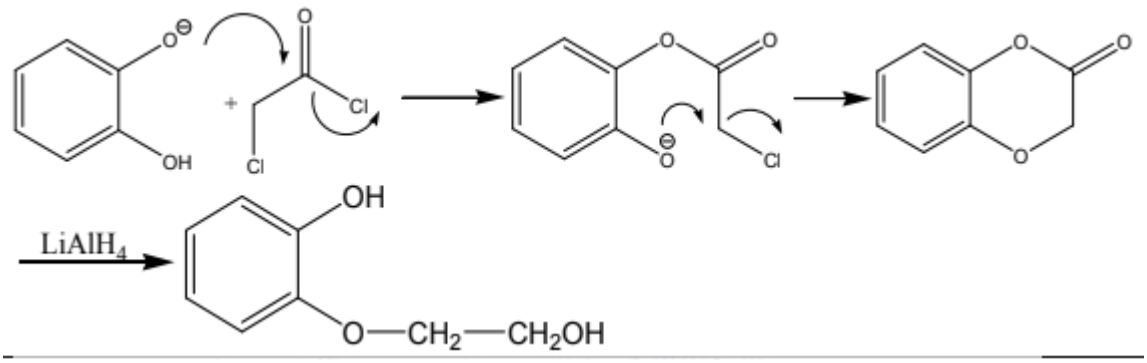


Q.2 Answer:

'Q' is



Solution:



Q.3 Answer:

Formaldehyde

Solution:

The composition of the organic compound is as follows:

$C = 40\%, H = 6.7\%$ **and** $O = 53\%$

$C : H : O = \frac{40}{12} : \frac{6.7}{1} : \frac{53}{16} = 3.33 : 6.7 : 3.33 = 1 : 2 : 1$

Empirical Formula is CH_2O

Empirical Weight $= 12 + 2 + 16 = 30$

V.D. $= 15$

Mol. weight $= 30$

Hence molecular formula is CH_2O.

Since, it reduces Fehling's solution,

Hence A **is an aldehyde.**

So it is formaldehyde (HCHO)

Q.4 Answer:

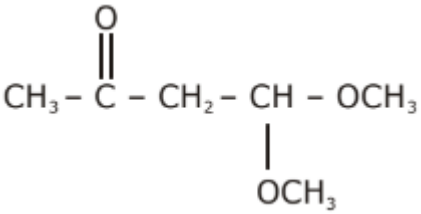
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$ClCH_2COOH$

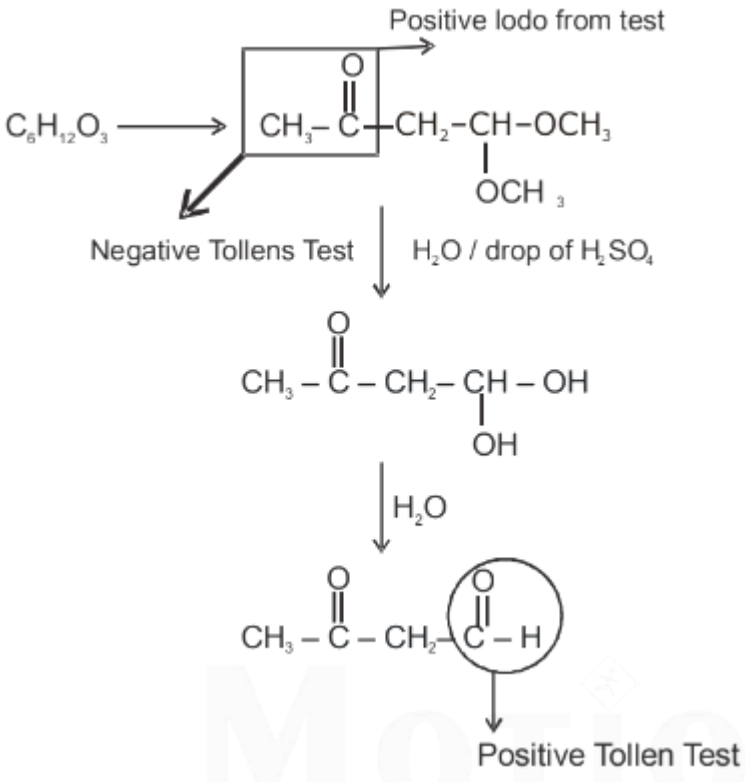
Solution:

Chloroacetic acid will be the strongest acid and as such its ionisation in aqueous solution will be more pronounced. Hence it will show minimum pH.

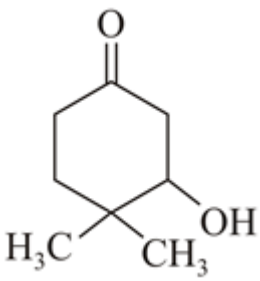
Q.5 Answer:



Solution:



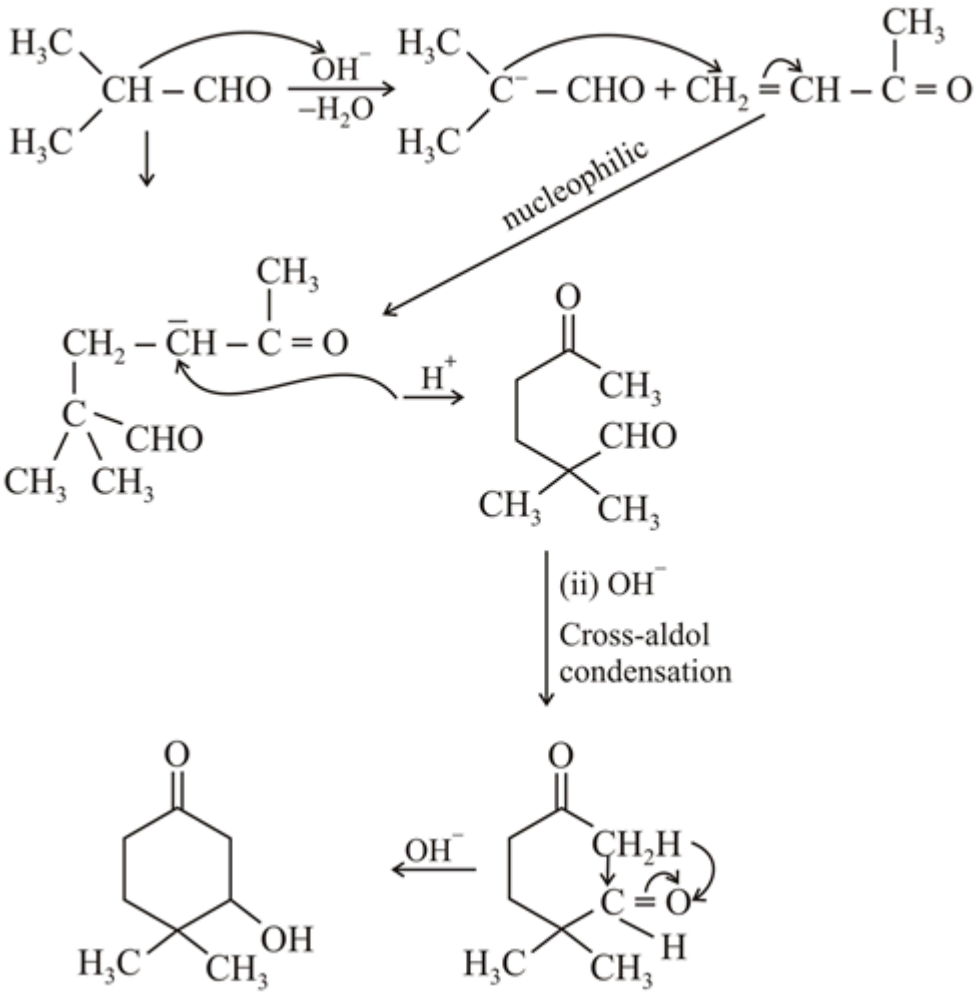
Q.6 Answer:



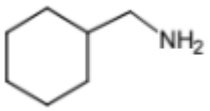
Solution:

The nucleophilic addition of carbanion of α, β-unsaturated carbonyl compounds is known as Michael addition

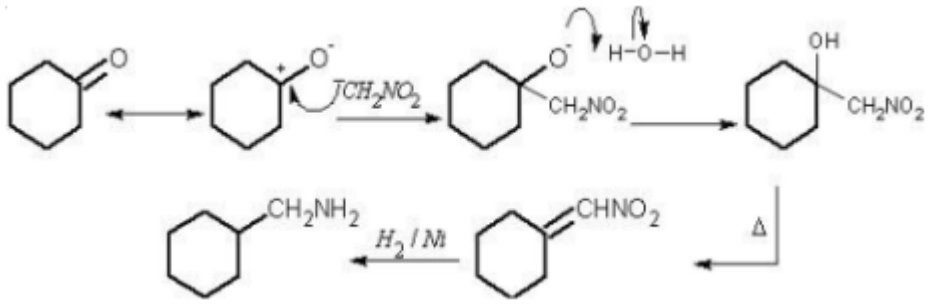
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Q.7 Answer:



Solution:



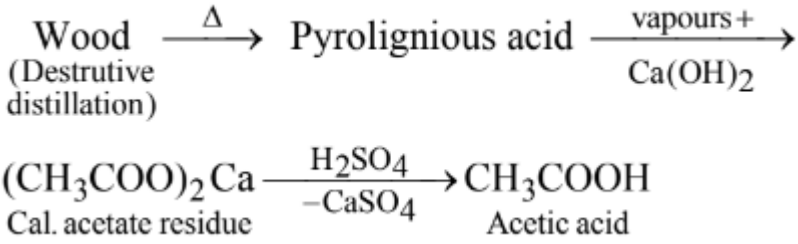
Q.8 Answer:

Acetic acid

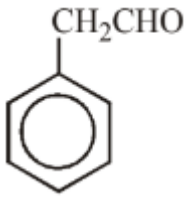
Solution:

Pyrolignious acid obtained by destructive distillation of wood can be used for the manufacture of the acetic acid by passing through $Ca(OH)_2$

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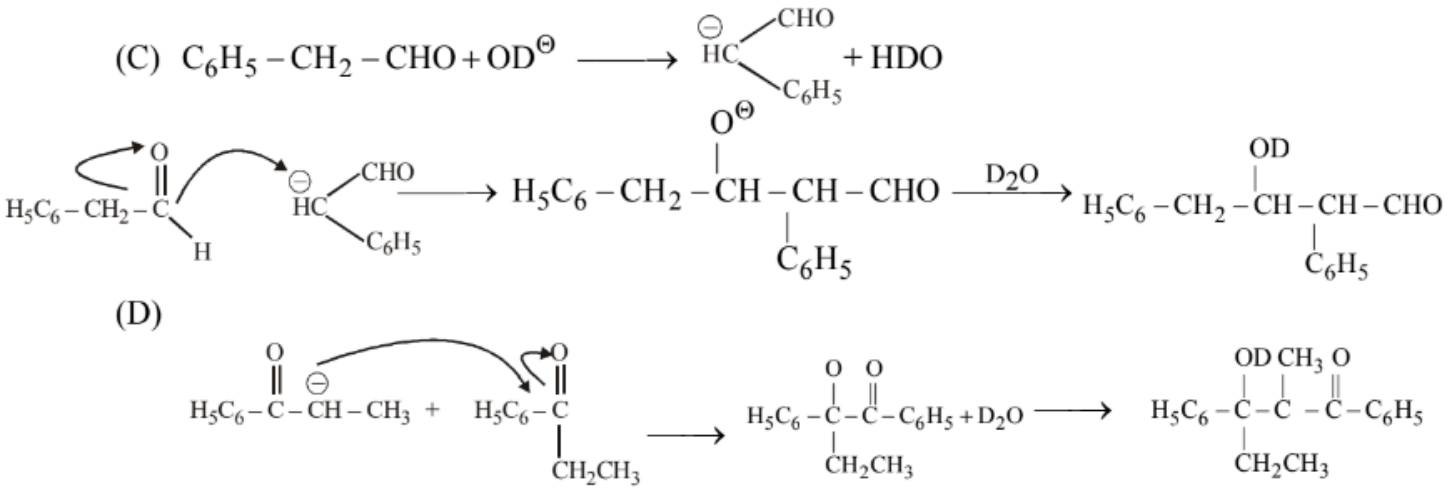


Q.9 Answer:



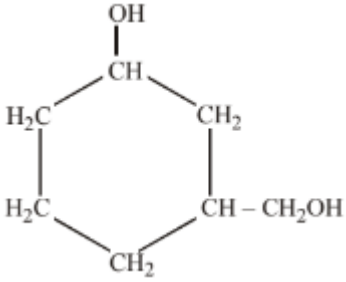
Solution:

A, B do not undergo aldol condensation, due to absence of α-hydrogen. C, D can undergo aldol condensation as below :



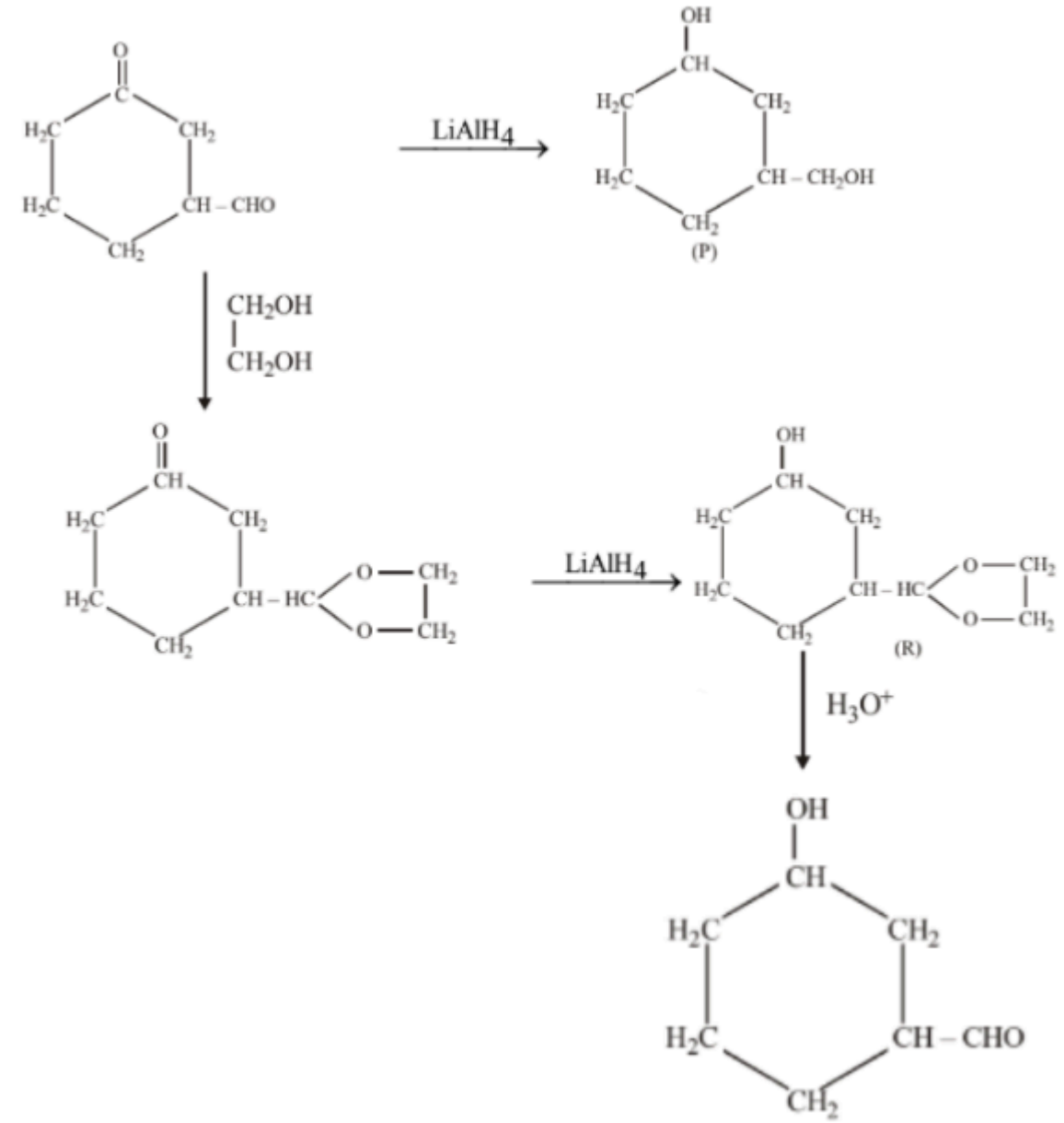
Q.10 Answer:

Structure P is



Solution:

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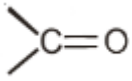
(S)

Q.11 Answer:

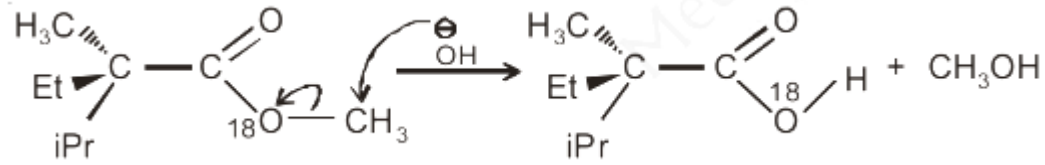
The given reaction primarily follows S_N2 mechanism

Solution:

In this reaction, O_H does not attack on



due to steric hinderance and it attacks on the back side of O^{18} and abstracts CH_3. This is the S_N2 mechanism



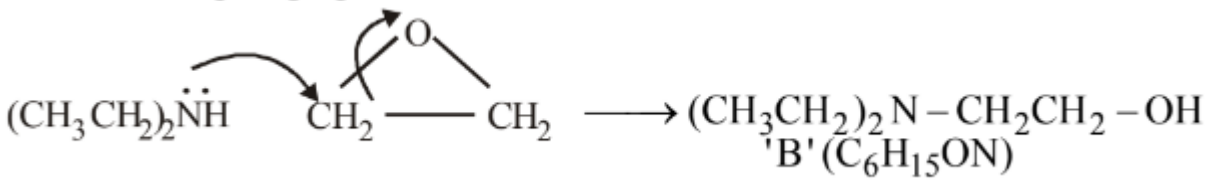
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Q.12 Answer:

amino acetate

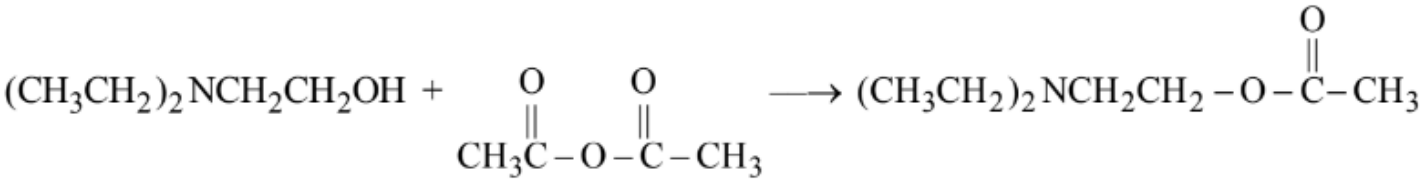
Solution:

(i) From the last paragraph



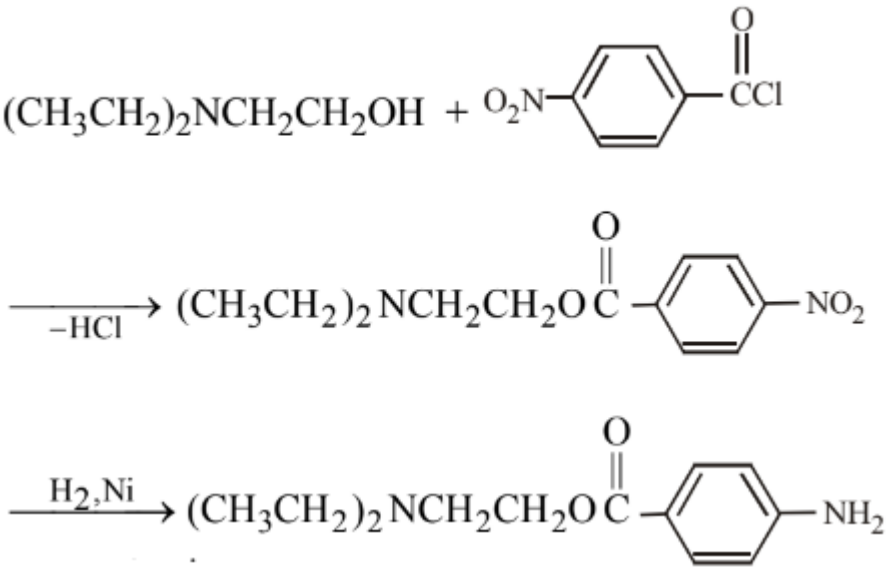
From the second last paragraph.,

$B+$ **acetic anhydride** $C \longrightarrow C\left(C_8H_{17}O_2N\right)$



Hence, C **is amino acetate.**

(ii)



(iii) Novocaine is an acid derivative

Novocaine $= A + B$

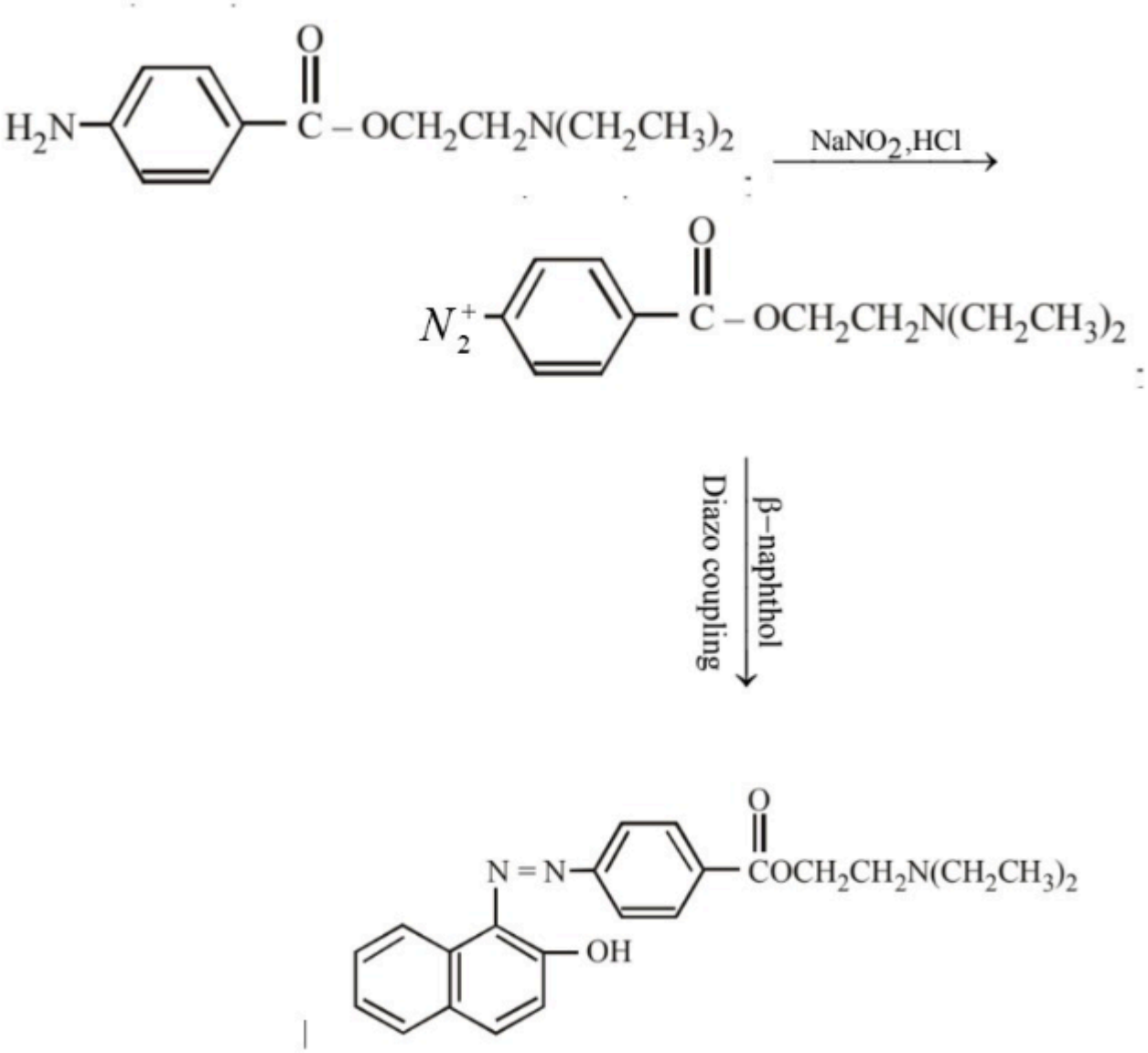
$B = (CH_3CH_2)_2NCH_2CHOH\,(C_6H_{16}ON)$; **Amino alcohol**

$A = C_2H_7O_2N$

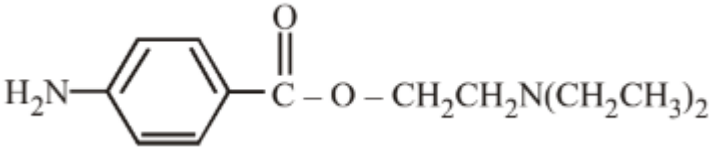
Therefore, in the light of other given information A **is aminobenzoic acid. As a hydrolysis of novocaine**

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occurs, thus, p-aminobenzoic acid.

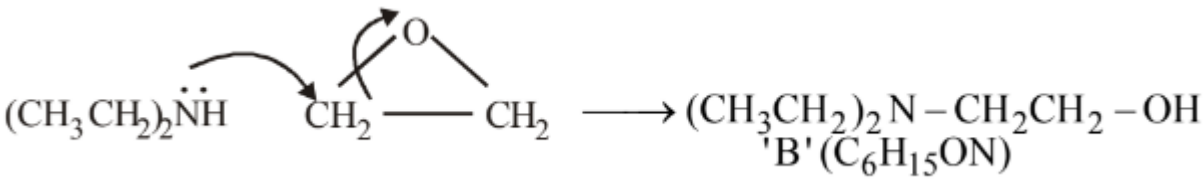


Q.13 Answer:



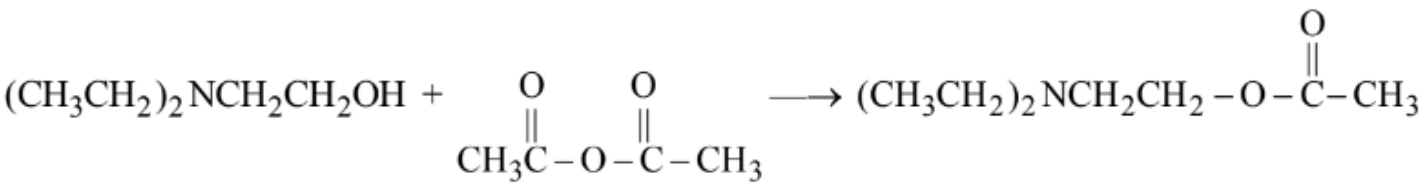
Solution:

(i) From the last paragraph



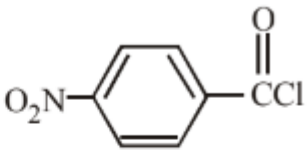
From the second last paragraph.,

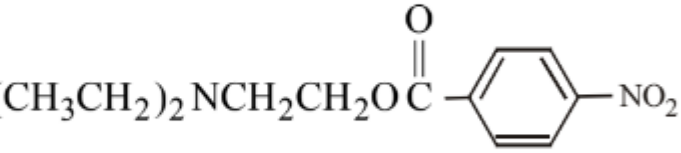
B+ **acetic anhydride** C $\longrightarrow$ C $(C_8H_{17}O_2N)$

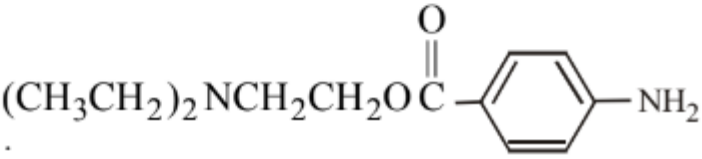


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Hence, C is amino acetate.

(ii) $(CH_3CH_2)_2NCH_2CH_2OH$ + 

$\xrightarrow[-HCl]{}$ 

$\xrightarrow{H_2,Ni}$ 

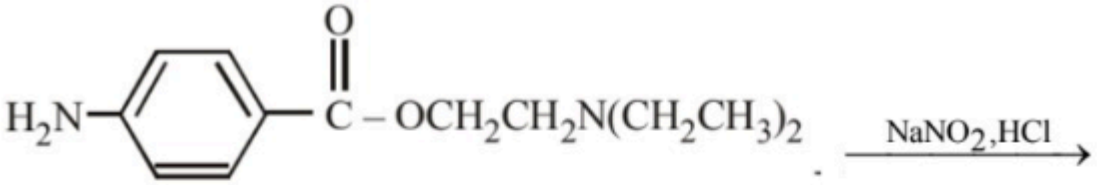
(iii) Novocaine is an acid derivative

Novocaine $= A + B$

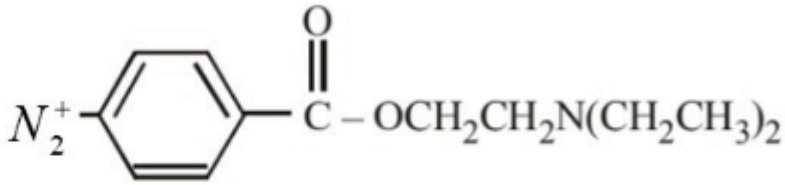
$B = (CH_3CH_2)_2\,NCH_2CHOH\,(C_6H_{16}ON)$; **Amino alcohol**

$A = C_2H_7O_2\,N$

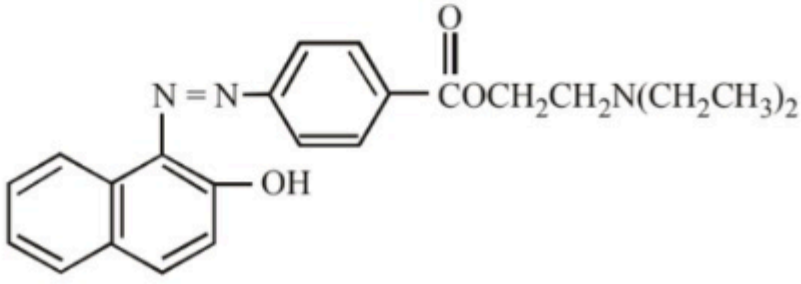
Therefore, in the light of other given information A is aminobenzoic acid. As a hydrolysis of novocaine occurs, thus, p-aminobenzoic acid.



$\xrightarrow{NaNO_2,HCl}$

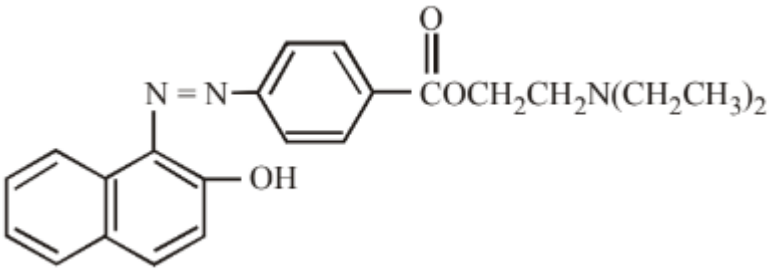


Diazo coupling, β–naphthol ↓



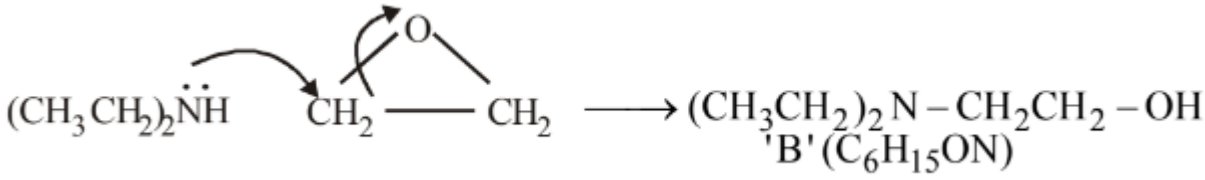
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Q.14 Answer:



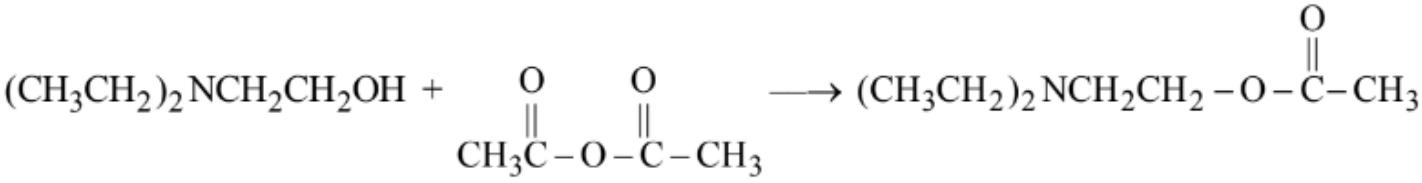
Solution:

(i) From the last paragraph



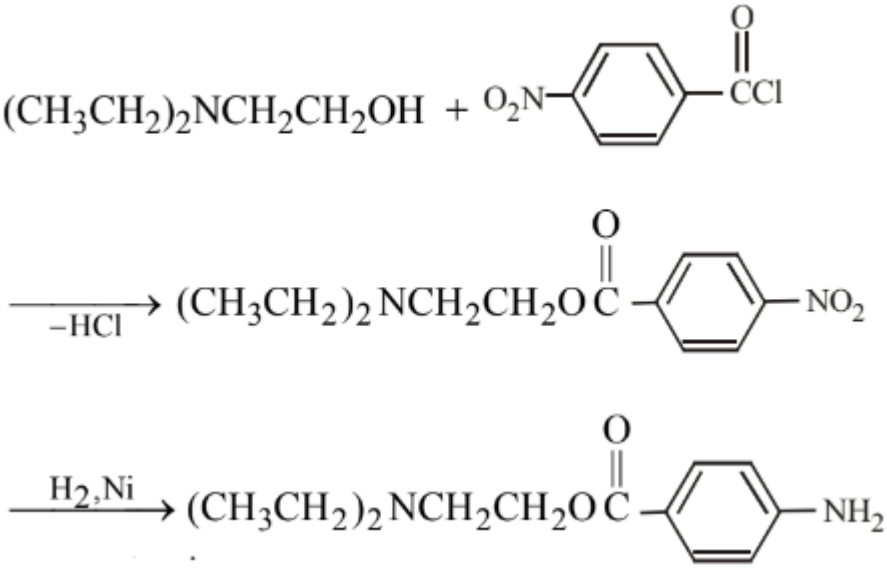
From the second last paragraph.,

$B+$ **acetic anhydride** $C \longrightarrow C\,(C_8H_{17}O_2\,N)$



Hence, C **is amino acetate.**

(ii)



(iii) Novocaine is an acid derivative

Novocaine $= A + B$

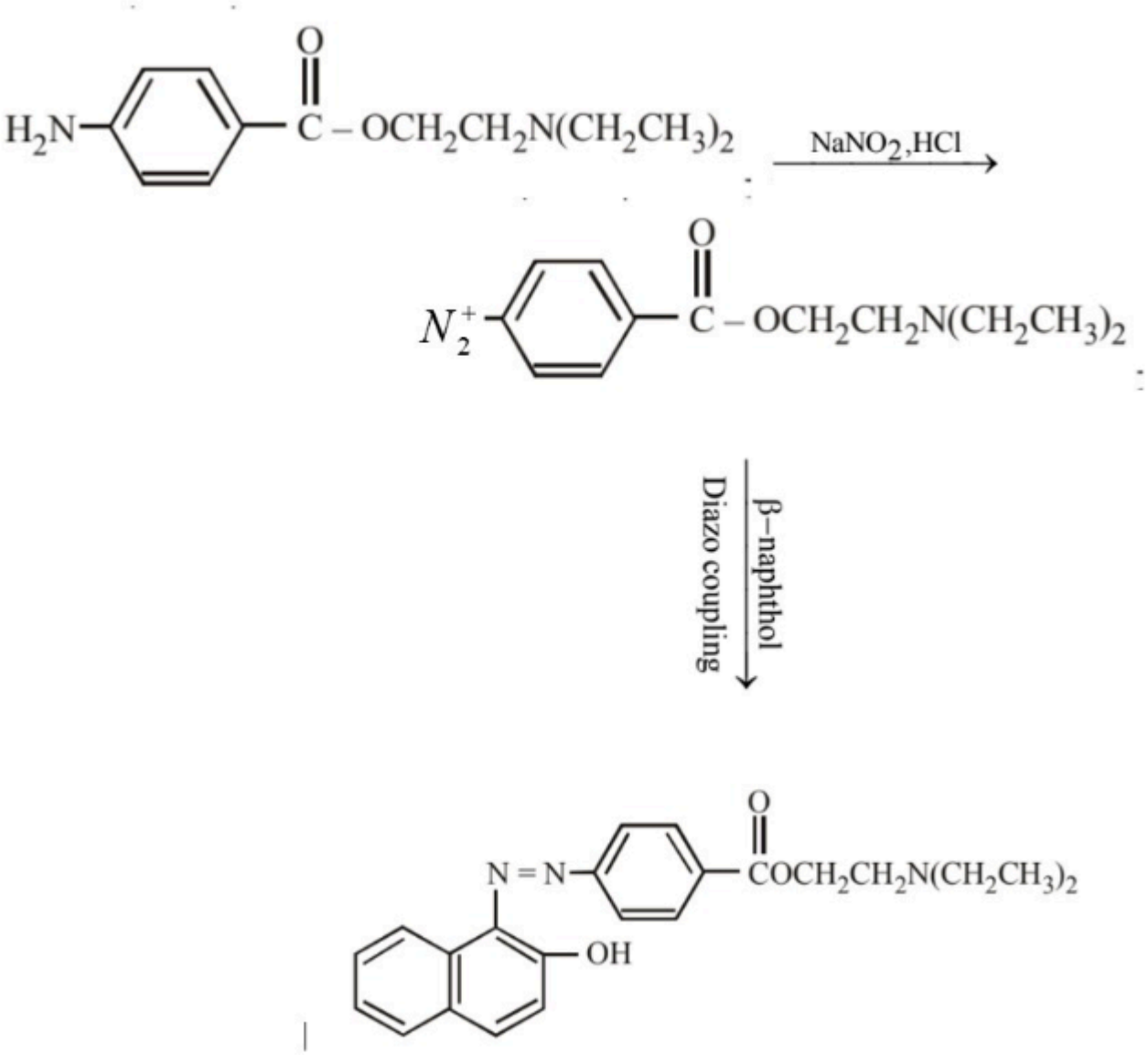
$B = (CH_3CH_2)_2\,NCH_2CHOH\,(C_6H_{16}ON))$; **Amino alcohol**

$A = C_2H_7O_2\,N$

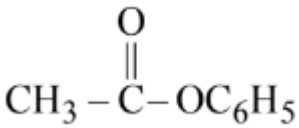
Therefore, in the light of other given information A **is aminobenzoic acid. As a hydrolysis of novocaine**

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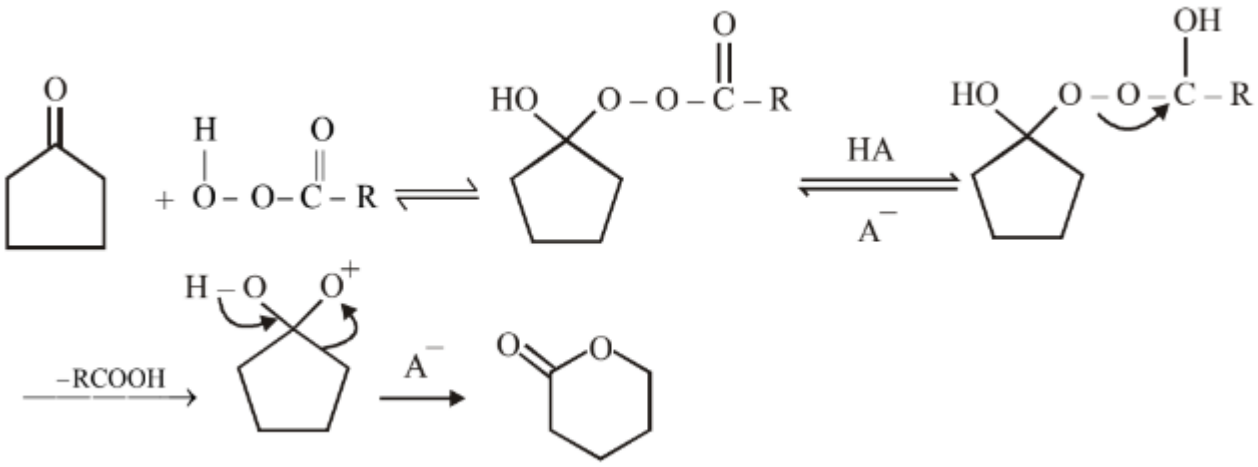
occurs, thus, p-aminobenzoic acid.



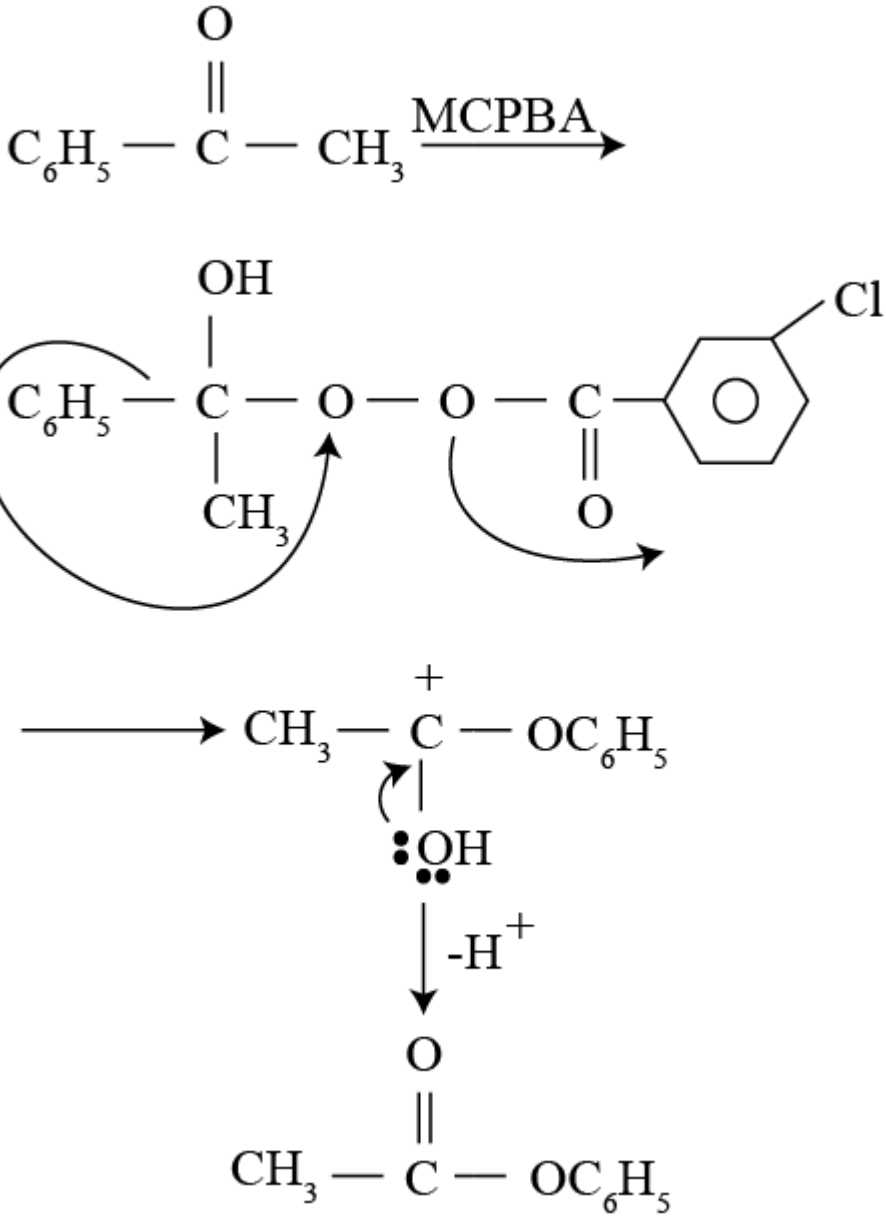
Q.15 Answer:



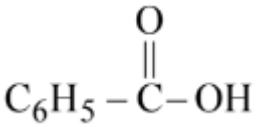
Solution:



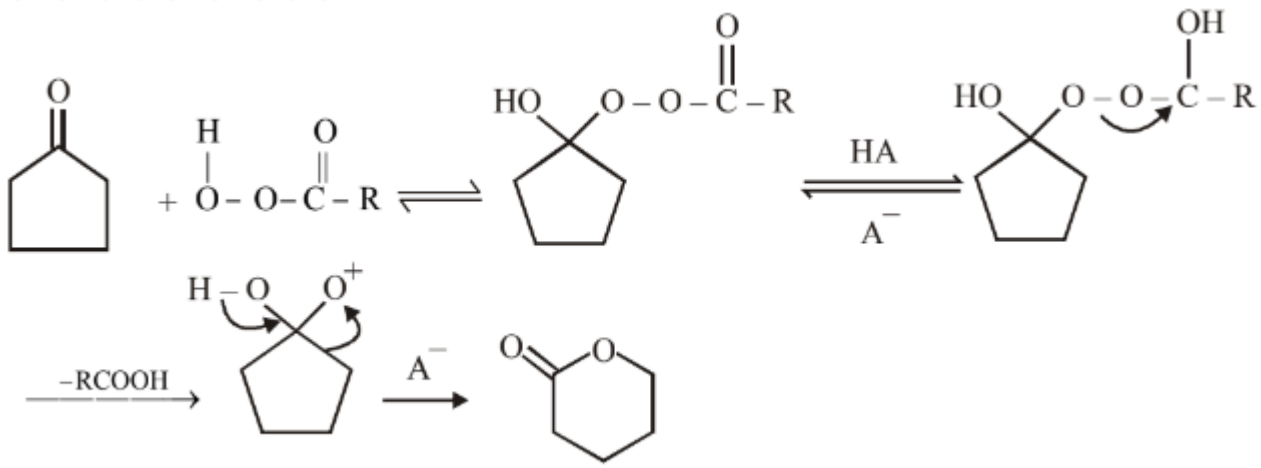
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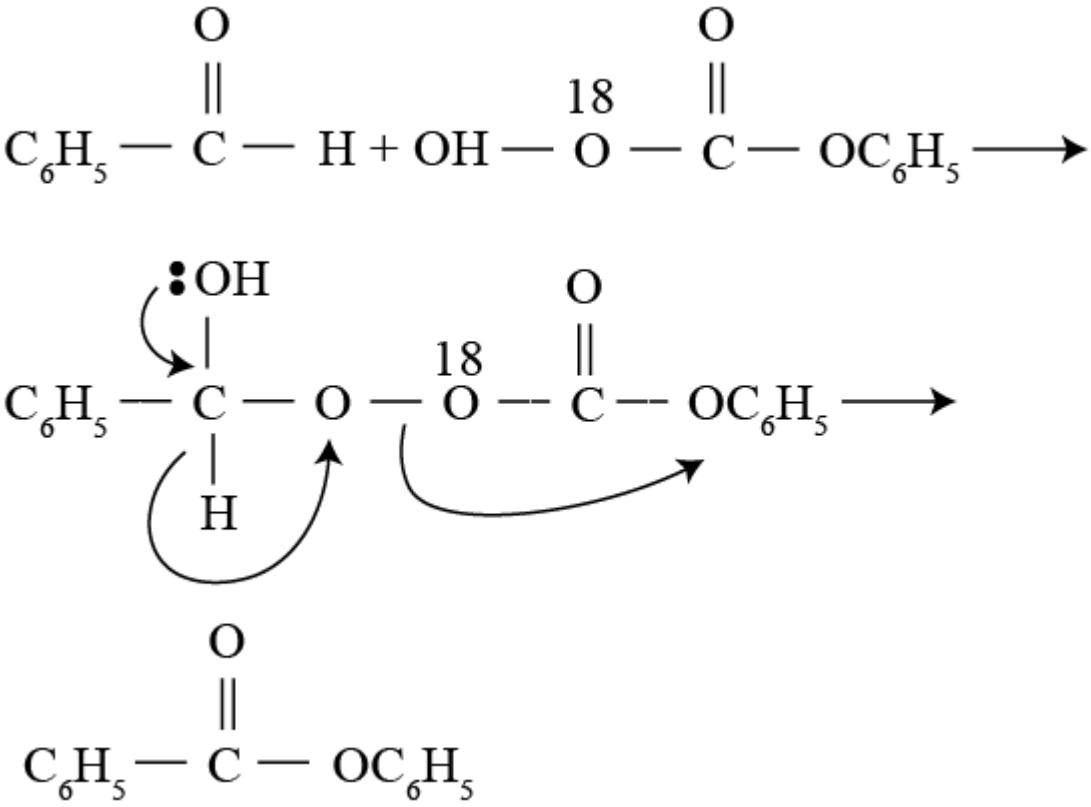
Q.16 Answer:



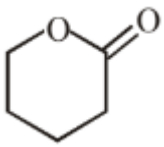
Solution:



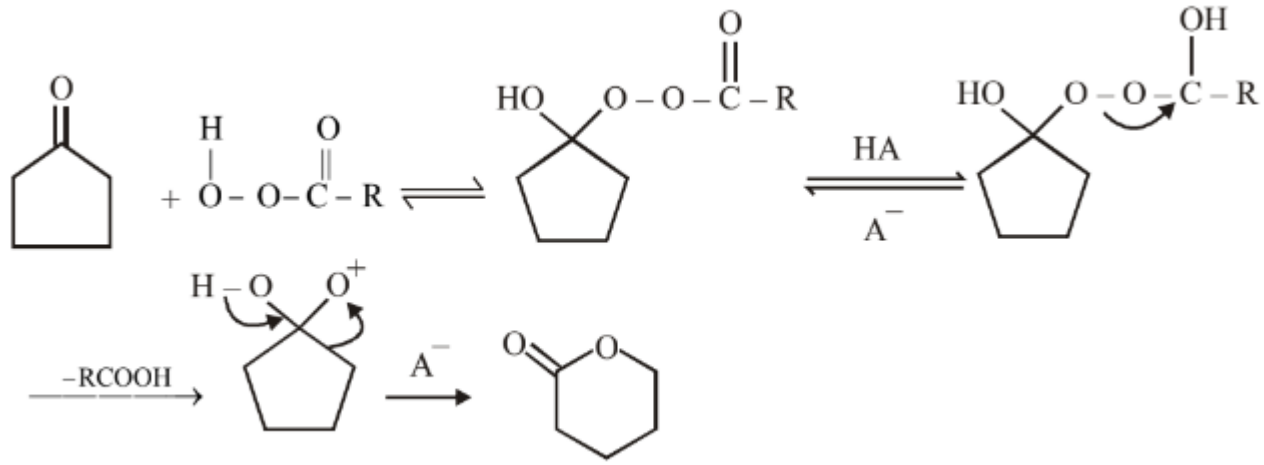
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Q.17 Answer:



Solution:



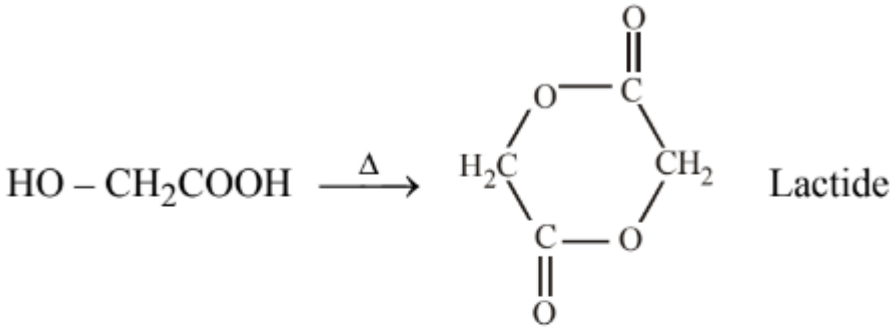
Q.18 Answer:

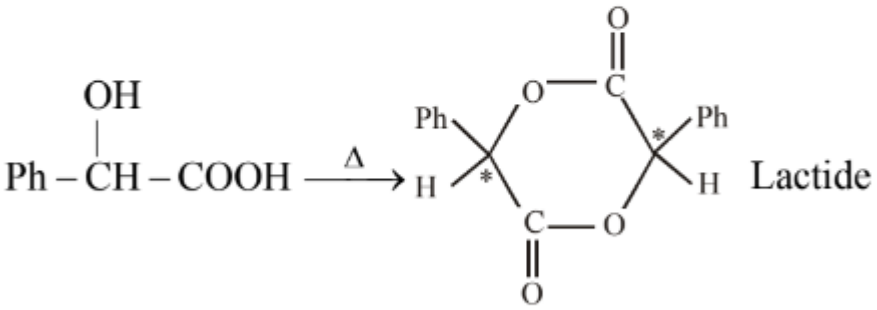
(A) - p,q, (B) - p,q,r,t, (C) - r, (D) - q,s,t,

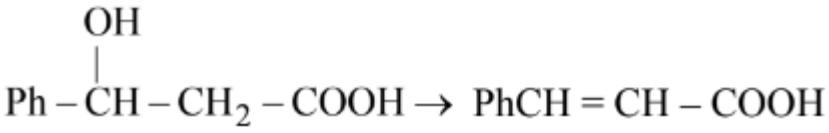
Solution:

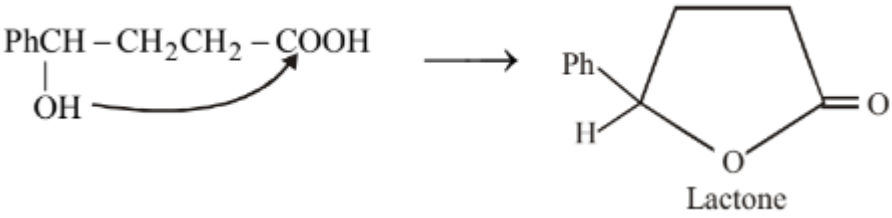
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(A) - pq, (B) - pqrt, (C) - r, (D) - qst.





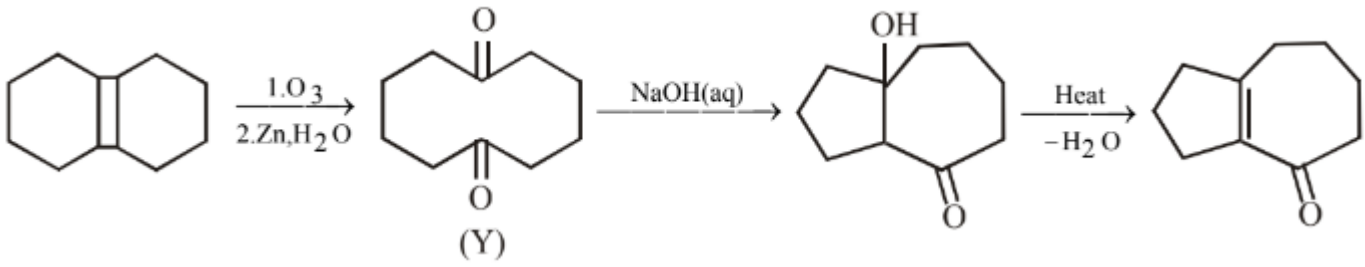




Q.19 Answer:

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Solution:



Q.20 Answer:

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Solution:

Let a gm of C_2H_5OH and bgm of CH_3CHO are present in the mixture

$a + b = 5.35$ .....**(1)**

$CH_3CHO + 2CuO \rightarrow CH_3COOH + Cu_2O$

red ppt.

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$143.6\ \mathrm{g Cu_2O}$ **is given by** $44\ \mathrm{g CH_3CHO}$

$0.12\ \mathrm{g}$ **of** $\mathrm{Cu_2O}$ **is given by** $\frac{44}{13.6} \times 0.12 = 0.0368\ \mathrm{g CH_3CHO}$

Thus, $\mathrm{b} = 0.0368\ \mathrm{g}$

$\% \mathrm{b} = \frac{0.368}{0.535} \times 100 = 6.868\% = 7\%$

The number of intramolecular aldol condensation products (α, β**-unsaturated carbonyl compound) formed from (Y) is 1.**

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